PUV Nearshore Wave Measurement Pipeline

Processing, Validation, and Spectral Bias Investigation

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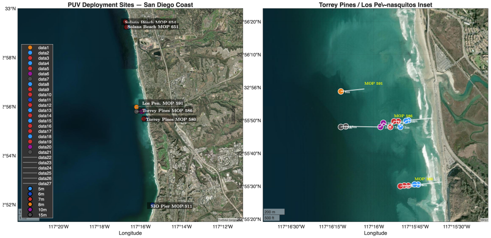
Outline

Motivation

- Nearshore wave transformation governs beach morphology, sediment transport, and coastal hazards
- Accurate in-situ wave measurements at 5–15 m depth are needed to validate spectral wave models and drive transport calculations
- PUV (pressure–velocity) measurements from bottom-mounted ADVs provide directional spectra without surface moorings
- Goal: build a robust, reproducible processing pipeline from raw Nortek Vector data to validated wave statistics

Instrument Network

- **20 deployments**, **40 instruments**
- 5 sites along San Diego coast:
  - Torrey Pines — MOP 580 & 586
  - SIO Pier — MOP 511
  - Solana Beach — MOP 654
  - Los Peñasquitos Lagoon — MOP 591
- Depths: 5–15 m
- Deployment period: 2023–2026
- Nortek Vector ADVs, 2 Hz continuous



Pipeline Architecture

1. **L1 — Raw to QC**
   Ingest burst files → clock drift correction → quality control → tilt correction → coordinate rotation
2. **L2 — Spectral Analysis**
   Segmentation → Welch PSD / cross-spectra → pressure correction → bulk parameters, bed velocity, stresses
3. **Validation**
   PUV vs. CDIP MOP wave model, spectral bias investigation
4. **Analysis scripts**
   Cross-deployment diagnostics, hypothesis testing

GitHub: `holdenlesliebole/Nortek_Vector_PUV_Pipeline`

Raw Data Ingestion

- Nortek Vector burst files: `.dat` (data), `.sen` (sensor/diagnostics), `.hdr` (header)
- Parsed with MATLAB `textscan` for speed — deployments can exceed 10^8 samples
- Continuous 2 Hz time series reconstructed from burst boundaries
- **Clock drift correction**: linear interpolation between deployment and recovery clocks
- Battery cutoff detection: gaps $> 1\ s \Rightarrow$ discard remainder of record

Quality Control — Old vs. New Approach

Old: Fixed threshold

- Absolute $\pm 5^\circ$ pitch/roll cap
- Discards all tilted data
- Problematic when pipes bend during storms — stable but non-zero tilt is *correctable*

New: Variability-based QC

- Rolling std of pitch/roll over 60 s windows
- Flag samples where variability $> 2^\circ$
- Absolute cap retained at 30° for severely compromised instruments
- Stable tilts *retained* and corrected

Tilt Correction

Sample-by-sample 3D rotation using measured pitch (ϕ) and roll (θ):

$$\mathbf{R} = \mathbf{R}_{\text{roll}}(\theta)\ \mathbf{R}_{\text{pitch}}(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\phi & 0 & \sin\phi \\ 0 & 1 & 0 \\ -\sin\phi & 0 & \cos\phi \end{bmatrix}$$

- Applied *before* heading rotation to buoy coordinates
- Corrects velocity components for instrument tilt
- Critical for instruments with bent mounting pipes (post-storm)

Coordinate Rotation

- Instrument XYZ $\rightarrow$ geographic buoy coordinates
- Convention: $+x$ West, $+y$ North, $+z$ Up (left-handed, MOP-consistent)
- Heading from deployment notes + magnetic declination via IGRF-13 model at deployment location and epoch
- Shore-normal rotation applied later in L2 using CDIP THREDDS shore-normal angle for each MOP station

L1 Processing Results

33 / 40 instruments processed successfully (**82.5%**)

7 failures — all hardware-related:

- Instrument burial (sand accretion)
- Bent or broken mounting pipes
- Dead / depleted batteries (no usable data)

No failures attributable to the processing pipeline itself.

Segmentation

- **Segment length**: 2048 samples $= 17.1$ min at 2 Hz
- Non-overlapping segments
- **NaN policy**: segments with $>$10% NaN discarded; $\leq$10% filled by linear interpolation
- Segment-mean pressure $\rightarrow$ water depth H
- All channels linearly detrended before spectral analysis

Spectral Estimation

- **Method**: Welch's averaged periodogram
- Sub-segments: $N_{\text{fft}} = 256$ samples (128 s)
- **Window**: Hanning, 50% overlap
- $\approx$**34 effective DOF** per segment
- Frequency resolution: $\Delta f \approx 0.0078\ Hz$, range 0–1 Hz
- **Cross-spectra**: S_{pu}, S_{pv} computed with same windowing for directional coefficients a_1, b_1

Pressure Correction

Surface elevation spectrum from pressure:

$$S_\eta(f) = \frac{S_p(f)}{K_p^2(f)}, \qquad K_p(f) = \frac{\cosh(k\, z_s)}{\cosh(k\, H)}$$

- Wavenumber $k(f)$: linear dispersion relation $\omega^2 = gk \tanh(kH)$ solved via Newton's method
- **Cutoff**: $K_p < 0.1 \Rightarrow$ set $S_\eta(f) = 0$ (noise suppression)
- Resulting f_{cut} recorded per segment
- Verified: Newton, Wu (1986), and O'Reilly formulations produce identical K_p (see Validation section)

Bulk Wave Parameters

- $H_s = 4\sqrt{m_0}$, computed separately for:
  - Sea–swell: 0.04–0.25 Hz
  - Infragravity: 0.004–0.04 Hz
- Peak period $T_p = 1/f_p$ (max S_η in SS band)
- Mean period $T_{m02} = \sqrt{m_0/m_2}$
- Mean direction from a_1, b_1 (PUV cross-spectra)
- Energy flux $F = \rho g \int c_g(f)\, S_\eta(f)\, df$

Additional L2 Products

- **Near-bed velocity**: IFFT method — scale velocity FFT by $\cosh(kH)/\cosh[k(H - z_s)]$, inverse transform
- **Bed shear stress**: Swart (1974) friction factor, $k_s = 2.5\, D_{50}$
- **Reynolds stress**: $\overline{u'w'}$, $\overline{v'w'}$, $\overline{u'v'}$; turbulent KE
- **Velocity moments**: skewness S_k, asymmetry A_s (Elgar 1985; Hoefel & Elgar 2003)

Shore-Normal Rotation

- Shore-normal angle retrieved from CDIP THREDDS for each MOP station
- Velocities rotated: $+x$ onshore, $+y$ alongshore north
- **Fallback**: if THREDDS unavailable, retain geographic buoy coordinates ($+x$ W, $+y$ N)
- Shore-normal frame required for cross-shore energy flux and transport calculations

MOP Wave Model

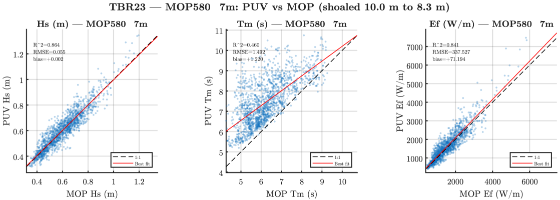
- CDIP Monitoring and Prediction (MOP) spectral wave model
- Initialized by offshore CDIP buoy observations
- Provides $S_\eta(f)$ at 10 m depth, hourly
- **Shoaling to PUV depth**:

$$S_\eta(f,\, h_{\mathrm{PUV}}) = S_\eta(f,\, 10\,\mathrm{m}) \times \frac{c_g(f,\, 10\,\mathrm{m})}{c_g(f,\, h_{\mathrm{PUV}})}$$

- PUV segments (17 min) interpolated to MOP hourly time base for comparison

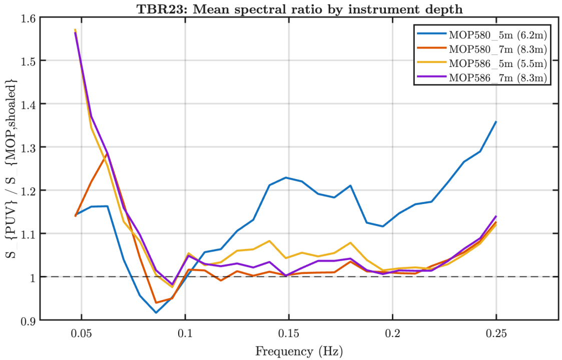
Bulk Comparison Results

- H_s: $R^2 = 0.83$–0.86
- RMSE $< 0.06\ m$
- Small positive bias (PUV $>$ MOP)
- Kelp-fouled instrument: $R^2 = 0.66$, bias $= +0.037\ m$



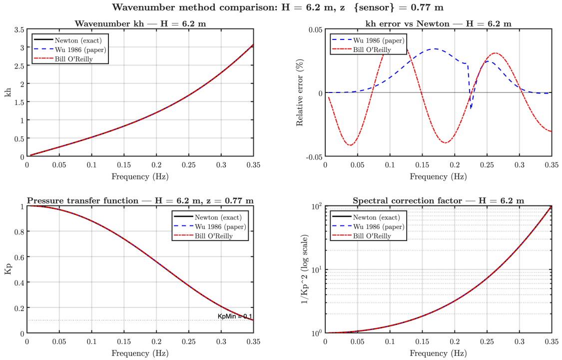
Frequency-Dependent Bias

- Excess energy concentrated at **swell peak** (0.04–0.08 Hz)
- $K_p > 0.9$ at these frequencies — pressure correction is minimal
- Mid-swell band: near-unity ratio (0.99–1.03)
- **NOT** a high-frequency pressure correction artifact



Pressure Correction Investigation

- Compared three wavenumber formulations:
  1. Newton's method (baseline)
  2. Wu (1986) approximation
  3. O'Reilly coefficients
- All produce **identical** K_p
- Pressure correction is **not** the source of the observed bias



Overview: Four Hypotheses

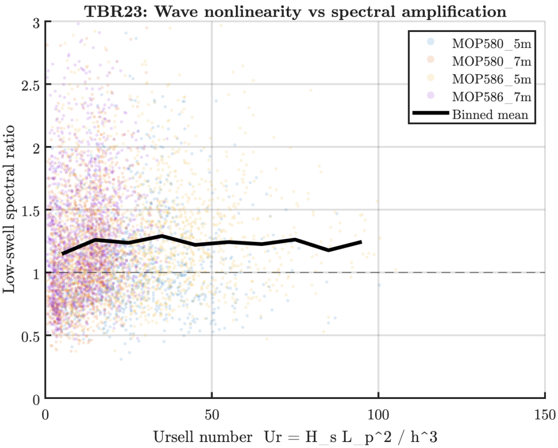
What causes the systematic positive H_s bias (PUV $>$ MOP)?

1. **Nonlinear shoaling** — triad interactions (Ursell number)
2. **Directional narrowing** — refraction-driven focusing
3. **Bound long waves** — group-forced IG contamination
4. **MOP spectral peak broadening** — numerical diffusion in the wave model

H1: Nonlinear Shoaling

- Test: correlate spectral ratio with Ursell number $Ur = H_s L_p^2 / h^3$
- $N = 5228$ segment-hours
- **Result**: $R^2 = 0.004$
- Median Ur: 12 at 8.3 m, 30 at 5.5 m

$\Rightarrow$ RULED OUT



H2: Directional Narrowing

- Refraction should *narrow* directional spread during shoaling $\Rightarrow$ energy focusing $\Rightarrow$ higher H_s
- **Result**: PUV spread is systematically *wider* than MOP at swell frequencies (15° vs. 10°)
- Opposite to prediction
- Correlation with H_s amplification: $R^2 < 0.001$

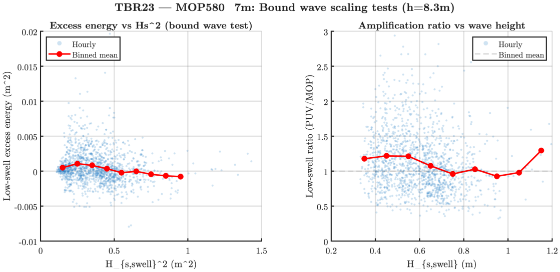
$\Rightarrow$ RULED OUT

Wider PUV spread may reflect scattered energy at shallow depth not captured by the forward-propagating MOP model.

H3: Bound Long Waves

- Bound wave energy $\propto H_s^2$ (Longuet-Higgins & Stewart 1962)
- **Result**: no positive correlation with H_s^2 ($R = -0.14$)
- Excess in *wrong frequency band* — at swell peak (0.04–0.08 Hz), not at group frequencies (0.01–0.03 Hz)

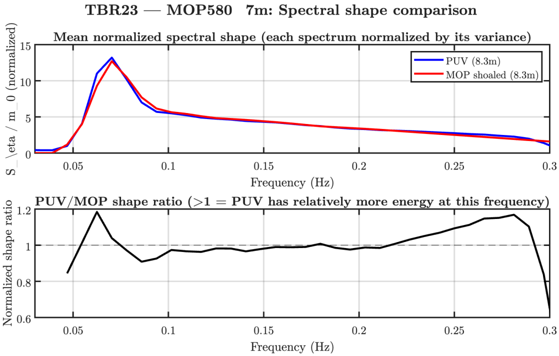
$\Rightarrow$ RULED OUT



H4: MOP Spectral Peak Broadening

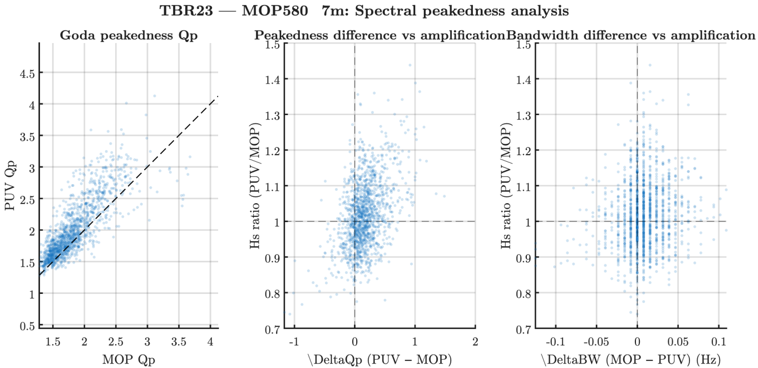
- Q_p (Goda peakedness): PUV 7% higher than MOP
- Half-power bandwidth: PUV **16.7% narrower**
- Peak spectral density: PUV 14.4% higher
- ΔQ_p vs. H_s amplification: $R = 0.44$, $R^2 = 0.20$

$\Rightarrow$ **SUPPORTED**



MOP model introduces numerical broadening of the spectral peak during propagation; PUV captures the true, narrower peak shape.

Peakedness Diagnostic



Hours when PUV spectrum is more peaked relative to MOP correspond to hours with larger positive H_s bias.

Hypothesis Summary

Hypothesis	**Test metric**	**Result**	**Verdict**
1. Nonlinear shoaling	$R^2(Ur, \text{ratio})$	$R^2 = 0.004$	Ruled out
2. Directional narrowing	PUV vs. MOP spread	PUV wider	Ruled out
3. Bound long waves	$R(H_s^2, \text{excess})$	$R = -0.14$	Ruled out
4. Peak broadening	$R(\Delta Q_p, \text{bias})$	$R = 0.44$	**Supported**

Cross-Deployment Confirmation

- Pattern confirmed across **11 instruments** from multiple deployments
- Correlation $R(Q_p) = 0.30$–0.63 across instruments
- Spans:
    - Summer and winter wave climates
    - Depths 5.5–15.5 m
    - Multiple MOP transects
- Spectral peak broadening is a **robust, general** feature of the MOP model — not site- or season-specific

L3 Architecture

- **L3a**: Frequency-band energy decomposition (IG / swell / sea)
- **L3b**: Storm event detection with MOP gap-filling
- **L3c**: Transport proxies (F_b, Shields, Rouse, mobilization)
- **L3d**: Tidal current decomposition (t_tide + NOAA depth)
- All components verified: E_f_total/L2.E_f = 1.0000 for all 33 instruments

Seasonal Energy Partitioning

- **Winter**: swell-dominated (65–90% of energy flux)
- **Summer**: sea-dominated or equal ($\sim$50/50)
- **Spring**: transitional
- Consistent across all 5 sites

Storm Detection + MOP Gap-Filling

- NN24 Dec 28, 2023: H_s up to 3.6 m, 74-hour duration
- MOP fills data gaps from instrument damage during storms
- Example: MOP651_7m detected 0 PUV storms, MOP found 3 (86-hour event)
- Critical for characterizing the forcing that damaged the instruments

Transport Proxy Depth Dependence

- Shields parameter decreases offshore: 0.24 (5.6 m) to 0.065 (15.5 m)
- Mobilization: 100% at 5 m, 78% at 15 m (winter)
- Rouse number: bedload-dominated across all sites
- Cumulative F_b: winter $3\times$ summer at equivalent depth

Tidal Validation

- NOAA Scripps Pier tide gauge: $R = 0.995$ with PUV depth
- Confirms PUV clocks are UTC (lag $= 0.0$ hours)
- Subtidal cross-shore current: consistently offshore (undertow)
- Tidal currents small ($<$0.05 m/s) vs. wave-driven flows

Key Findings

1. **Pipeline validated**: 33/40 instruments processed (82.5%), strong agreement with MOP ($R^2 > 0.83$, RMSE < 0.06 m)
2. **Bias identified**: small systematic positive H_s bias concentrated at the swell spectral peak, *not* from the pressure correction
3. **Mechanism determined**: MOP model introduces numerical spectral peak broadening; PUV captures the true narrower peak ($R = 0.44$ between ΔQ_p and bias)
4. **Cross-deployment robustness**: pattern confirmed across 11 instruments, summer/winter, 5.5–15.5 m depth
5. **Full L1–L3 pipeline operational**: all 33 instruments processed through forcing characterization, storm detection, and transport proxy computation

Implications for TBR23 Paper

- Pipeline is validated and ready for production use
- Bias is understood and attributable to the model, not the measurement — no correction needed for PUV data
- Spectral shape finding *supports* using PUV as ground truth for nearshore wave characterization
- L2 products (bed velocity, stress, moments) are ready for sediment transport analysis

Next Steps

- **Cross-deployment analysis**: expand when CDIP THREDDS recovers (currently intermittent)
- **IG / bound-wave quantification**: separate forced vs. free IG energy
- **Tidal decomposition**: segment analysis by tidal phase and water level
- **Grain size integration**: laser diffraction D_{50} for site-specific bed roughness
- **L3 transport calculations**: combine wave statistics with morphology data
- **Potential spinoff paper**: wave spectral shape evolution during shoaling — novel PUV-based quantification of model spectral broadening

Questions?